



GIRIJANANDA CHOWDHURY UNIVERSITY, GUWAHATI
DEPARTMENT OF CIVIL ENGINEERING

Formative Assessment - I

Geotechnical Engineering (B. Tech 4th Semester)

Course: BCE23403T

Batch: 2023-2027

25th February 2025

Duration: 1 hour

Academic Year: 2024-2025

Total Marks: 30

	Questions	Marks	CO	Cognitive Level
1	a) Define C_u and D_{30} . b) What do MH, GW and SC mean? c) Define Seepage velocity and Discharge velocity d) Explain any one clay mineral.	$3 \times 4 = 12$	1	Understand
2	A constant head permeability test was run on a sand sample 16 cm in length and 60 cm^2 in cross-sectional area. Porosity was 40%. Under a constant head of 30 cm, the discharge was found to be 45 cm^3 in 18 seconds. Calculate the coefficient of permeability. Also determine, the discharge velocity and seepage velocity during the test.	6	2	Apply
3	The unit weight of a sand backfill was determined by field measurements to be 1746 kg/m^3 . The water content at the time of test was 8.6 % and the unit weight of the solid constituents was 2.6 gm/m^3 . In the laboratory, the void ratios in the loosest and densest states were found to be 0.642 and 0.462 respectively. What was the relative density of the fill?	6	2	Apply
4	A horizontal stratified soil deposit consists of 3 uniform layers of thickness 6, 4 and 12 m respectively. The permeability of these layers are $8 \times 10^{-4} \text{ cm/s}$, $52 \times 10^{-4} \text{ cm/s}$ and $6 \times 10^{-4} \text{ cm/s}$, find the effective average permeability of the deposit in the horizontal and vertical direction.	6	2	Apply